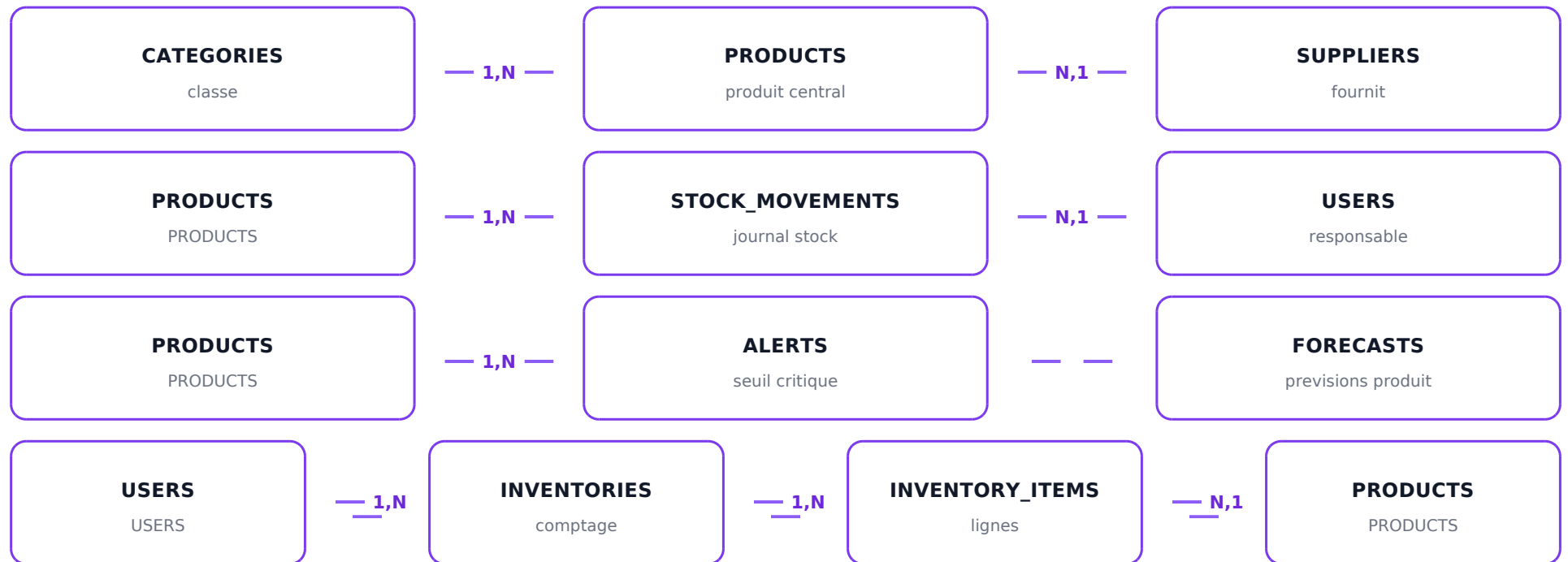
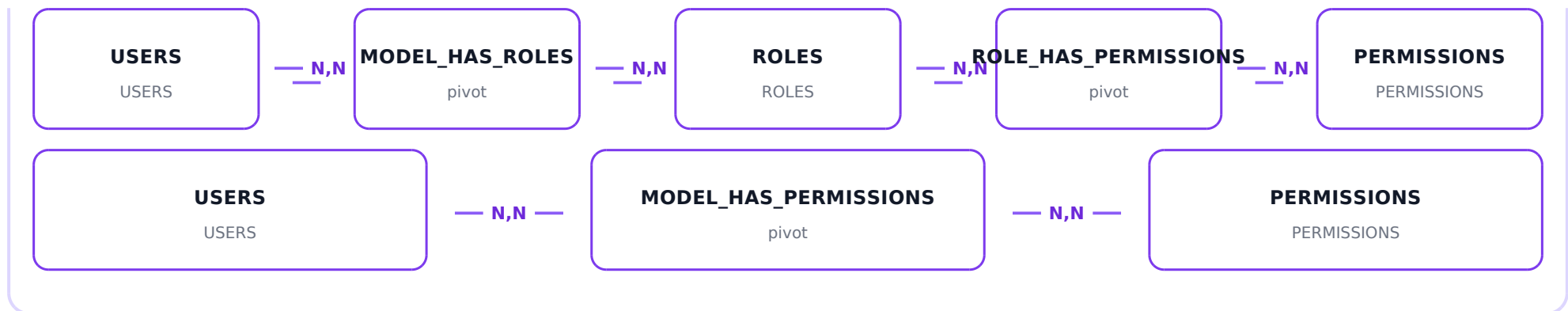


Schema de base de donnees - StockFlow

Vue simplifiee : uniquement les tables et les liaisons. Les champs sont detailles apres le diagramme.

Lecture : les boites sont les tables, les connecteurs indiquent la cardinalite principale. Les pivots Spatie et les tables associatives metier sont affichees comme des tables a part entiere.





Relations N,N a retenir

- **users N,N roles** via `model_has_roles`.
- **roles N,N permissions** via `role_has_permissions`.
- **users N,N permissions** via `model_has_permissions`.
- **inventories N,N products** via `inventory_items`.
- **products N,N users** via `stock_movements`, relation historique/evnementielle.

Dictionnaire des tables

users				
Utilisateurs - Comptes applicatifs et statut d activation.				
CHAMP	TYPE	CLES / CONTRAINTES		DEFAULT
id	INTEGER	PK	NOT NULL	-
name	varchar	NOT NULL		-
email	varchar	NOT NULL		-
email_verified_at	datetime	-		-
password	varchar	NOT NULL		-
status	varchar	NOT NULL		'active'
last_login_at	datetime	-		-
remember_token	varchar	-		-
created_at	datetime	-		-
updated_at	datetime	-		-

categories

Categories - Referentiel de classement des produits.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
name	varchar	NOT NULL	-
description	TEXT	-	-
status	varchar	NOT NULL	'active'
created_at	datetime	-	-
updated_at	datetime	-	-

suppliers

Fournisseurs - Referentiel des fournisseurs.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
name	varchar	NOT NULL	-
contact_name	varchar	-	-
phone	varchar	NOT NULL	-
email	varchar	-	-
address	TEXT	-	-
status	varchar	NOT NULL	'active'
created_at	datetime	-	-
updated_at	datetime	-	-

products

Produits - Catalogue produit et stock courant.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
category_id	INTEGER	FK categories.id NOT NULL	-
supplier_id	INTEGER	FK suppliers.id NOT NULL	-
name	varchar	NOT NULL	-
sku	varchar	NOT NULL	-
barcode	varchar	-	-
purchase_price	numeric	NOT NULL	-
sale_price	numeric	NOT NULL	-
current_stock	numeric	NOT NULL	'0'
critical_stock	numeric	NOT NULL	-
unit	varchar	NOT NULL	-
status	varchar	NOT NULL	'active'
created_at	datetime	-	-

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
updated_at	datetime	-	-

stock_movements

Mouvements de stock - Journal des entrees, sorties et ajustements.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
product_id	INTEGER	FK products.id NOT NULL	-
user_id	INTEGER	FK users.id NOT NULL	-
type	varchar	NOT NULL	-
quantity	numeric	NOT NULL	-
stock_before	numeric	NOT NULL	-
stock_after	numeric	NOT NULL	-
reference	varchar	-	-
notes	TEXT	-	-
movement_date	datetime	NOT NULL	-
created_at	datetime	-	-
updated_at	datetime	-	-

inventories

Inventaires - Sessions de comptage physique.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
user_id	INTEGER	FK users.id NOT NULL	-
inventory_date	date	NOT NULL	-
status	varchar	NOT NULL	'draft'
notes	TEXT	-	-
validated_at	datetime	-	-
created_at	datetime	-	-
updated_at	datetime	-	-

inventory_items

Lignes d inventaire - Produits comptes dans un inventaire, avec ecart.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
inventory_id	INTEGER	FK inventories.id NOT NULL	-
product_id	INTEGER	FK products.id NOT NULL	-
expected_quantity	numeric	NOT NULL	-
actual_quantity	numeric	NOT NULL	-
difference	numeric	NOT NULL	-
created_at	datetime	-	-
updated_at	datetime	-	-

alerts

Alertes - Historique des alertes de seuil critique.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
product_id	INTEGER	FK products.id NOT NULL	-
type	varchar	NOT NULL	'critical_stock'
message	TEXT	NOT NULL	-
status	varchar	NOT NULL	'new'
triggered_at	datetime	NOT NULL	-
resolved_at	datetime	-	-
created_at	datetime	-	-
updated_at	datetime	-	-

forecasts

Previsions - Historique des calculs de rupture et recommandations.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
product_id	INTEGER	FK products.id NOT NULL	-
average_daily_usage	numeric	NOT NULL	-
predicted_out_date	date	-	-
recommended_quantity	numeric	NOT NULL	-
risk_score	INTEGER	NOT NULL	-
generated_at	datetime	NOT NULL	-
created_at	datetime	-	-
updated_at	datetime	-	-

roles

Roles - Roles Spatie Permission.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
name	varchar	NOT NULL	-
guard_name	varchar	NOT NULL	-
created_at	datetime	-	-
updated_at	datetime	-	-

permissions

Permissions - Permissions Spatie Permission.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
id	INTEGER	PK NOT NULL	-
name	varchar	NOT NULL	-
guard_name	varchar	NOT NULL	-
created_at	datetime	-	-
updated_at	datetime	-	-

model_has_roles

Attribution roles - Pivot entre utilisateurs et roles.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
role_id	INTEGER	PK FK roles.id NOT NULL	-
model_type	varchar	NOT NULL	-
model_id	INTEGER	NOT NULL	-

model_has_permissions

Permissions directes - Pivot entre utilisateurs et permissions directes.

CHAMP	TYPE	CLES / CONTRAINTES	DEFAULT
permission_id	INTEGER	PK FK permissions.id NOT NULL	-
model_type	varchar	NOT NULL	-
model_id	INTEGER	NOT NULL	-

role_has_permissions

Permissions par role - Pivot entre roles et permissions.

CHAMP	TYPE	CLES / CONTRAINTES		DEFAULT
permission_id	INTEGER	PK	FK permissions.id	NOT NULL
role_id	INTEGER	FK roles.id	NOT NULL	-

Tables techniques Laravel

- **sessions** : table technique Laravel.
- **cache** : table technique Laravel.
- **cache_locks** : table technique Laravel.
- **jobs** : table technique Laravel.
- **job_batches** : table technique Laravel.
- **failed_jobs** : table technique Laravel.
- **password_reset_tokens** : table technique Laravel.